

Generation Interconnection Request System Impact Study Report

For Queue Project Q19-1274

Haskell Ave 138 kV
November 27, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1274. The Interconnection Customer proposes a 166.1 MW battery energy storage facility interconnecting at the Haskell Ave 138 kV substation in the Pennsylvania area. The request was received on August 25, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 440 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 13 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$90,264,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Haskell Ave to Seney Uppco 138 kV
Contingency: Acdia to Montour Sta 500 kV
Loading: 131.0% (Rating: 235 MVA)
DFAX: 17.3%
Pre-project loading: 118.8%
2. Facility: Haskell Ave to S Bessemer 138 kV
Contingency: Haskell Ave to O'Brien Substation 138 kV
Loading: 123.2% (Rating: 249 MVA)
DFAX: 80.3%
Pre-project loading: 69.7%
3. Facility: Jackie Howard to Montour Sta 138 kV L
Contingency: Hermanas 138/1kV Autotransformer
Loading: 113.9% (Rating: 267 MVA)
DFAX: 7.0%
Pre-project loading: 109.5%
4. Facility: Bowen St to S Bessemer 138 kV L
Contingency: Haskell Ave to Seney Uppco 138 kV
Loading: 113.2% (Rating: 227 MVA)
DFAX: 65.4%
Pre-project loading: 65.2%
5. Facility: Mauston to Long Beach 138 kV
Contingency: Westwood Generation Llc 345/138 kV Auto
Loading: 105.8% (Rating: 175 MVA)
DFAX: 5.7%
Pre-project loading: 100.4%

6. Facility: Bowen St to Rawhide 138 kV
Contingency: Westwood Generation Llc to Morganville 345 kV
Loading: 105.3% (Rating: 419 MVA)
DFAX: 21.6%
Pre-project loading: 96.7%
7. Facility: O'Brien to Haskell Ave 138 kV
Contingency: Haskell Ave to S Bessemer 138 kV
Loading: 104.6% (Rating: 247 MVA)
DFAX: 46.1%
Pre-project loading: 73.6%
8. Facility: O'Brien to Sta B Century 138 kV
Contingency: Haskell Ave to S Bessemer 138 kV
Loading: 103.1% (Rating: 240 MVA)
DFAX: 46.1%
Pre-project loading: 71.3%
9. Facility: Eklutna to Allegheny Hydro No 8 345 kV
Contingency: Piney Crk Project to Caldwell 345 kV
Loading: 103.0% (Rating: 1,188 MVA)
DFAX: 5.6%
Pre-project loading: 102.3%
10. Facility: Highland Cooper Tap to Maury 138 kV Line
Contingency: South Bessemer to Haskell Ave 138 kV
Loading: 102.4% (Rating: 189 MVA)
DFAX: 46.1%
Pre-project loading: 62.0%
11. Facility: Proler to Station B Century Substation 138 kV
Contingency: Haskell Ave to South Bessemer 138 kV
Loading: 101.4% (Rating: 234 MVA)
DFAX: 46.1%
Pre-project loading: 68.7%
12. Facility: Black Knight Substation to Mauston 138 kV Line
Contingency: Westwood Generation Llc 345 kV/138 kV Auto
Loading: 100.2% (Rating: 200 MVA)
DFAX: 5.7%
Pre-project loading: 95.5%
13. Facility: Fronton to Highland Cooper Tap 138 kV L
Contingency: Haskell Ave to South Bessemer 138 kV
Loading: 100.0% (Rating: 200 MVA)
DFAX: 46.1%
Pre-project loading: 61.7%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Seney Uppco Substation to Haskell Ave 138 kV	Rebuild line with higher-capacity conductor	\$52,805,000	100% (\$52,805,000)
Haskell Ave Substation to S Bessemer 138 kV line	Rebuild line with higher-capacity conductor	\$9,939,000	100% (\$9,939,000)
Jackie Howard to Montour Sta 138 kV L	Rebuild line with higher-capacity conductor	\$6,069,000	34% (\$2,043,000)
Bowen St to S Bessemer 138 kV L	Rebuild line with higher-capacity conductor	\$2,652,000	100% (\$2,652,000)
Mauston to Long Beach 138 kV	Reconductor line	\$5,025,000	100% (\$5,025,000)
Bowen St to Rawhide 138 kV	Reconductor line	\$1,127,000	72% (\$816,000)
Haskell Ave to O'Brien 138 kV	Reconductor line	\$1,201,000	100% (\$1,201,000)
O'Brien to Sta B Century 138 kV	Reconductor line	\$1,062,000	100% (\$1,062,000)
Eklutna to Allegheny Hydro No 8 345 kV	Reconductor line	\$14,943,000	11% (\$1,626,000)
Highlan Cooper Tap to Maury Substation 138 kV	Reconductor line	\$2,064,000	100% (\$2,064,000)
Proler to Station B Century Substation 138 kV	Reconductor line	\$2,726,000	100% (\$2,726,000)
Black Knight to Mauston 138 kV	Reconductor line	\$1,317,000	100% (\$1,317,000)
Fronton to Highland Cooper Tap 138 kV L	Reconductor line	\$6,988,000	100% (\$6,988,000)
	Total allocated to project		\$90,264,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.